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An Improved Equation for Predicting the Stress of Bonded High-Strength Strands at Flexural Failure

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Abstract

To achieve efficient design and ensure the safety of concrete structures, the use of high-strength concrete, reinforcing steel, and prestressing tendons has been steadily increasing. In this study, for flexural design of prestressed concrete (PSC) structures employing high-strength strands with tensile strengths of 2160 MPa and 2360 MPa, the applicability of the current design-code equation for predicting the strand stress at flexural failure (f_{ps})—which was originally proposed based on studies of conventional strands with tensile strengths of 1860 MPa or lower—was evaluated. Furthermore, an improved prediction equation was proposed. Section analyses based on stress–strain curves obtained from numerous tensile tests of high-strength strands were conducted, and the results were compared with the existing prediction equations specified in ACI 318 and the Korean KDS code. The comparison revealed that, for high-strength strands, the strand stress tends to be underestimated in the tension-controlled region and overestimated in the compression-controlled region. To address these issues, a new prediction equation was proposed that retains the form of the existing equation but incorporates correction factors reflecting the characteristics of high-strength strands. The performance of the proposed equation was evaluated not only for rectangular sections but also for T- and I-shaped sections, and its predictive accuracy was verified by comparing the predicted strand stresses and nominal flexural strengths with those obtained from section analyses. As a result, the proposed prediction equation demonstrated improved accuracy compared with the existing one, while maintaining an appropriate level of conservatism. Therefore, it is expected to enhance design efficiency for PSC structures employing high-strength strands.

Keywords: prestressed concrete; prestressing tendon; high-strength strand; strand stress; flexural strength

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1. Introduction

In response to the growing trend toward larger-scale, super high-rise, and long-span infrastructure, the use of high-strength concrete, reinforcing steel, and prestressing tendons in concrete structures has been increasing to achieve efficient design and ensure structural safety. While extensive research on high-strength concrete and reinforcing steel has been accumulated and is being put into practical use [1,2], studies on high-strength strands used in prestressed concrete (PSC) structures remain relatively limited. PSC is an effective technique for achieving longer spans, enhancing structural efficiency, and securing durability through crack control in concrete structures. As such, it has been widely applied to various concrete structures, including bridges, circular tanks, roofs, and large-scale

buildings. Recently, in an effort to further improve structural performance beyond that achieved with conventional strands having a tensile strength of 1860 MPa, there has been a growing trend toward developing and utilizing high-strength strands whose tensile strength exceeds this level. When high-strength strands are applied to structures, the prestressing force per strand can be increased, thereby enhancing the overall prestressing effect. Alternatively, an equivalent level of prestressing can be achieved even with fewer strands or with smaller or fewer anchorage devices. Consequently, high-strength strands offer advantages such as reducing the cross-sectional dimensions of PSC members, enabling longer spans, and decreasing material usage [3–5].

In particular, Korea has taken a leading role in the field of high-strength strands by incorporating 2160 MPa (SWPC7CL) and 2360 MPa (SWPC7DL) grade strands into the Korean Industrial Standards (KS) [6,7] ahead of international and foreign standards such as ASTM [8], EN [9], JIS [10], and ISO [11]. For the 2360 MPa grade, the initial specification established in 2011 was 2400 MPa; however, it was later adjusted to 2360 MPa to better align with international standards [6]. Foreign standards such as ASTM [8] and JIS [10] address only conventional strands with a tensile strength of 1860 MPa, while EN [9] mentions strands up to the 2160 MPa grade but imposes restrictions on the nominal diameters available. Meanwhile, ISO [11] has incorporated the 2160 MPa and 2360 MPa grades—alongside the conventional 1860 MPa grade—in alignment with Korea, reflecting the growing use of high-strength strands in Korea and other countries.

In Korea, the design and construction of long-span PSC girder bridges using 2360 MPa high-strength strands have been increasing in recent years. Meanwhile, in the United States, studies have been conducted on pretensioned PSC girders using Grade 300 strands (tensile strength of 2070 MPa) instead of the conventional Grade 270 strands (tensile strength of 1860 MPa) [12]. In Japan, 2230 MPa high-strength strands have been developed and applied to several bridge projects [13]. Although the use of high-strength strands in structural applications has been increasing worldwide and is beginning to be reflected in several standards, research verifying whether the equations validated and proposed for conventional strands of 1860 MPa or lower—currently adopted in design codes—are also applicable to high-strength strands remains insufficient. Even in Korea's KDS provisions, where high-strength strands are actively used, it is stated that when 2160 MPa or 2360 MPa high-strength strands are used in design, their applicability must be verified through testing or analysis [14]. However, because it is difficult for design practitioners to verify such applicability through actual testing or analysis, design equations developed for conventional strands are still frequently used for high-strength strands in practice. This may undermine the safety and serviceability of PSC members incorporating high-strength strands. Therefore, this study aims to evaluate the applicability of existing design provisions—originally proposed for conventional strands—through flexural analysis of PSC members using high-strength strands, and further to propose a prediction equation for strand stress at flexural failure that is suitable for high-strength strands.

2. Literature Review

One of the key design parameters in the flexural design of PSC members is the strand stress at flexural failure in the ultimate state, f_{ps}. The value of f_{ps} can be rigorously calculated through section analysis based on strain compatibility and force equilibrium conditions. However, in practical design, approximate equations for estimating f_{ps} provided in design codes are primarily used for the sake of computational simplicity.

Mattock [15] performed section analysis for rectangular sections with 1860 MPa bonded strands and proposed an f_{ps} prediction equation based on the results. However, for T-shaped sections rather than rectangular ones, when the number of strands

exceeds a certain level, the net tensile strain in the strands quickly reaches the compression-controlled strain limit. This occurs because the neutral axis moves out of the top flange and approaches the tension edge, increasing the influence of compression-controlled behavior and thereby raising the likelihood of brittle failure. To prevent this, Mattock specified a maximum reinforcement ratio of prestressing tendon to promote ductile failure, thereby ensuring safety in design. Currently, both the American ACI 318 [16] and the Korean KDS [14] provide f_{ps} prediction equations based on the formula proposed by Mattock, offering the advantage of a relatively simple application procedure.

On the other hand, Naaman et al. [17] and Loov [18] proposed prediction equations for 1860 MPa bonded strands that include the neutral axis depth as a variable. By taking the neutral axis depth into account, these equations offer the advantage of predicting strand stresses more accurately across various section shapes, unlike the formula proposed by Mattock [15]. However, these equations have the drawback of computational complexity, as the f_{ps} and neutral axis depth must be determined through the prediction equation and the force equilibrium equation. The CSA A23.3 [19] and AASHTO LRFD [20] design codes calculate f_{ps} based on the prediction equation proposed by Loov [18].

Since the existing prediction equations were derived based on 1860 MPa strands, their applicability to high-strength strands with higher tensile strength has not been sufficiently examined. Park et al. [21] evaluated the applicability of the prediction equations proposed by Mattock [15] and Loov [18] to 2160 MPa and 2360 MPa bonded high-strength strands. For I-shaped sections, a comparison between f_{ps} obtained from rigorous section analysis and that predicted by the equations showed that the f_{ps} calculated using the prediction equations was overestimated, resulting in non-conservative outcomes. Therefore, Park et al. [21] proposed two prediction equation models by distinguishing between ranges of strand strain. Specifically, the existing AASHTO LRFD [20] equation is used in the inelastic range, while a modified prediction equation with an added correction term is applied in other ranges, yielding conservative estimates of f_{ps}—lower than those from section analysis—while improving prediction accuracy. However, since this approach requires prior evaluation of strand strains during member design, it focuses more on improving accuracy rather than simplifying calculations, making it somewhat difficult and cumbersome to apply in practical design.

Therefore, in this study, to balance design convenience with relatively accurate prediction of high-strength strand stresses, the applicability of Mattock's [15] prediction equation—familiar to practitioners through American and Korean design codes—was first analyzed for high-strength strands. Based on this analysis, a correction equation reflecting the characteristics of high-strength strands is proposed.

3. Evaluation of the Applicability of Existing f_{ps} Prediction Equation Through Section Analyses

3.1. Overview of Existing Prediction Equations

In this study, to evaluate the applicability of the f_{ps} prediction equation proposed by Mattock [15] and adopted in ACI 318 [16] and KDS [14], shown in Equation (1), to high-strength strands, the results from section analyses were compared with those obtained from the prediction equation. Equation (1) is applicable only under the condition $f_{pe} \geq 0.5 f_{pu}$, where f_{pe} and f_{pu} are the effective prestress and the tensile strength of prestressing tendon, respectively. Section analysis refers to the process of accurately evaluating the flexural behavior of PSC members by using strain compatibility and force equilibrium conditions for sections in which strands and concrete act compositely [22]. Examples of such bonded strands include pretensioned members and post-tensioned members with grouted ducts. In the case of unbonded strands, strain compatibility does not hold, making the application of

such section analysis difficult. In section analysis, the strand stress at the ultimate state, f_{ps}, can be determined using an iterative method based on the strand strain at flexural failure and the strand stress–strain relationship. Therefore, by comparing the strand stresses obtained from section analysis with those calculated using Equation (1), the applicability of the existing prediction equation to high-strength strands was examined.

$$f_{ps} = f_{pu}\left[1 - \frac{\gamma_p}{\beta_1}\left\{\rho_p \frac{f_{pu}}{f_{ck}} + \frac{d}{d_p}(\omega - \omega')\right\}\right], \tag{1}$$

where f_{ps}: stress of prestressing tendon at nominal flexural strength, f_{pu}: specified tensile strength of prestressing tendon, f_{ck}: specified compressive strength of concrete, γ_p: factor for the type of prestressing tendon, β_1: factor for the depth of an equivalent rectangular compressive stress block, ρ_p: reinforcement ratio of prestressing tendon (=A_p/bd_p), A_p: area of prestressing tendon, b: width of section, d: effective depth of tension reinforcing bar, d_p: effective depth of prestressing tendon, ω: reinforcement index of tension reinforcing bar, and ω': reinforcement index of compression reinforcing bar. Also refer to Figure 1 for more information on the notations.

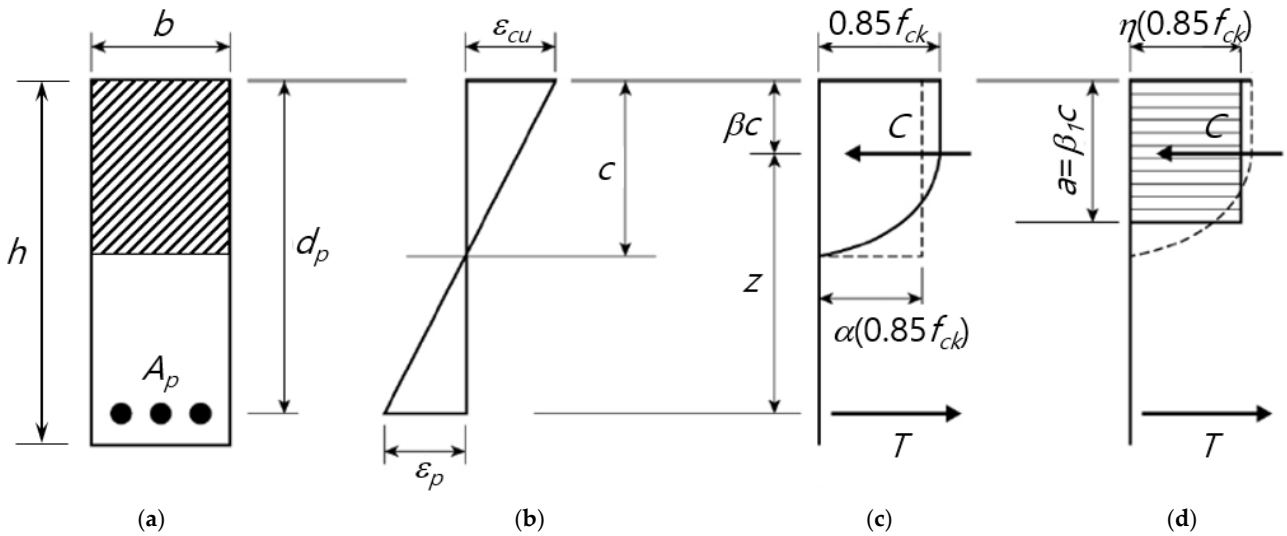

Figure 1. Ultimate state of a PSC beam: (**a**) beam section; (**b**) strain distribution; (**c**) exact stress distribution; and (**d**) equivalent approximate stress distribution.

3.2. Section Analysis Conditions and Material Models

To apply the strain compatibility and force equilibrium conditions in section analysis, the stress–strain relationship of the strands was first established. With reference to Figure 1, the strain distribution along the section depth was assumed to be linear according to beam theory, and the tensile resistance of concrete—that is, the tensile stress—was neglected in accordance with conventional assumptions. In addition, flexural failure was assumed to occur when the concrete compression fiber reached the ultimate strain, ε_{cu}. The section analysis was performed considering only the strands, following the methodology of previous studies, and the stress and force distributions of concrete and strands at the ultimate state, as shown in Figure 1, were based on the Korean KDS provisions [23,24], which reference and improve upon the corresponding provisions of Eurocode 2 [25].

Referring to previous studies [26] and design practice, the jacking stress in the strands was set to 70% ($f_{pj} = 0.7f_{pu}$), considering that the maximum allowable stressing level in design codes is approximately 80% of the tensile strength [14,16,24]. For prestress losses, an immediate loss ratio of 10% ($f_{pi} = 0.9f_{pj}$) and an effective prestress level of 85% ($f_{pe} = 0.85f_{pi}$) were applied to account for both immediate and time-dependent losses.

In addition, the specified compressive strength of concrete (f_{ck}) used in the analysis was set to 40 MPa. A preliminary review indicated that varying the strand stress level or the concrete specified compressive strength had little influence on the subsequent section analysis results; thus, these assumptions were adopted. Meanwhile, the concrete material model determining the shape of the compressive stress distribution in Figure 1 followed the Korean design codes [23,24]. Therefore, ε_{cu} was taken as 0.0033, corresponding to f_{ck} = 40 MPa. Although this differs slightly from the value of 0.003 used by Mattock [15] and ACI 318 [16], regardless of f_{ck}, the difference was found to have no significant impact on the results.

For the strand material model, the Ramberg–Osgood model, as expressed in Equation (2) and Figure 2, was applied, which is known to reasonably approximate the actual strand stress–strain behavior. According to KS [6], strands are classified as Type B, C, or D based on their tensile strength, with tensile strengths of 1860 MPa, 2160 MPa, and 2360 MPa, respectively. Type B corresponds to conventional-strength strands, while Types C and D are considered high-strength strands. Table 1 presents the constant values that determine the form of the mathematical equation. For Type B strands, previously proposed values were used [21,22,26], while for high-strength strands, values were derived through regression analysis based on tensile test results reported in related studies [27]. The numbers of specimens of high-strength strands were 15 and 39 for Types C and D, respectively.

$$f_p = E_p \varepsilon_p \left[A + \frac{1 - A}{\left\{ 1 + \left(B\varepsilon_p \right)^C \right\}^{1/C}} \right] \leq f_{pu}, \tag{2}$$

where f_p: stress of prestressing tendon, E_p: modulus of elasticity of prestressing tendon (=200,000 MPa), ε_p: strain of prestressing tendon, and A, B, and C: constants related to the shape of stress–strain curve, shown in Table 1. The Ramberg–Osgood model derived in this regression exhibits very high accuracy compared to the tensile test data, with average coefficients of determination (R^2) of 0.996 and 0.995 for Types C and D, respectively. As shown in Figure 2 and Table 1, high-strength strands exhibit distinct differences from conventional-strength strands not only in tensile and yield strengths but also in yield strain and the overall shape of the stress–strain curve. Therefore, it is necessary to examine whether the existing f_{ps} prediction equation, derived based on the stress–strain curve of Type B strands, can also appropriately predict the behavior of high-strength strands, by considering the stress–strain curves and section analysis of Type C and D strands.

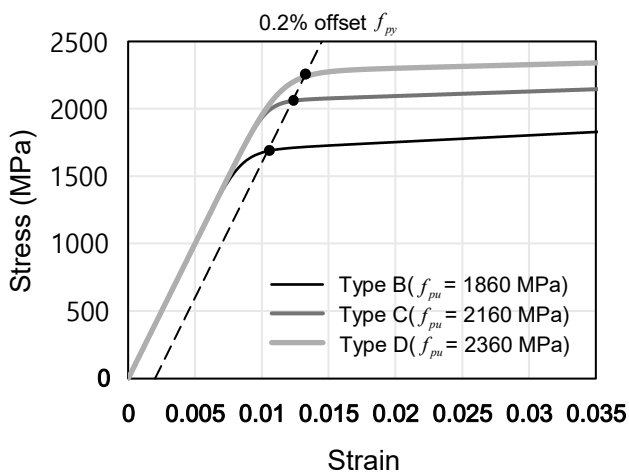

Figure 2. Stress–strain curves for different strand types.

Table 1. Constants and yield properties of the Ramberg–Osgood model for different strand types.

Strand Type	Tensile Strength, f_{pu} (MPa)	Constant A	Constant B	Constant C	0.2% Offset Yield Strength, f_{py} (MPa)	Yield Strain, ε_{py}
Type B	1860	0.0250	118.0	10.0	1685	0.0104
Type C	2160	0.0169	96.97	14.71	2059	0.0123
Type D	2360	0.0131	87.72	9.98	2238	0.0132

3.3. Strain and Stress of Strands at Flexural Failure

Given the stress–strain relationship of the strands, the strand strain at flexural failure can be determined through section analysis, and the corresponding strand stress, f_{ps}, can be calculated. For bonded strands, f_{ps} typically satisfies $f_{py} < f_{ps} < f_{pu}$, whereas for unbonded strands, f_{ps} is often less than f_{py} due to the averaging of strand strains. Figure 3 illustrates the procedure for determining the strain of bonded strands at flexural failure [22]. At the stage where only prestressing force acts on the section, ① represents the state after immediate and time-dependent prestress losses, where the strand strain corresponds to ε_1 in Equation (3). As the applied load increases, the compressive strain in the concrete at the strand location reaches zero, entering a decompression state as shown in ②, and the corresponding increase in strand strain is denoted as ε_2 in Equation (4). As the load continues to increase, cracks develop at the bottom of the section, and the neutral axis rises toward the compression fiber. When the compressive strain in the concrete reaches the ultimate strain, ε_{cu}, as shown in ③, the section reaches the failure state. The corresponding increase in strand strain at this stage is denoted as ε_3 in Equation (5). Therefore, the total strand strain at flexural failure, ε_{ps}, can be calculated as the sum of these strains, as given in Equation (6). The calculated ε_{ps} can then be substituted into Equation (2), representing the stress–strain relationship of high-strength strands, to determine f_{ps}.

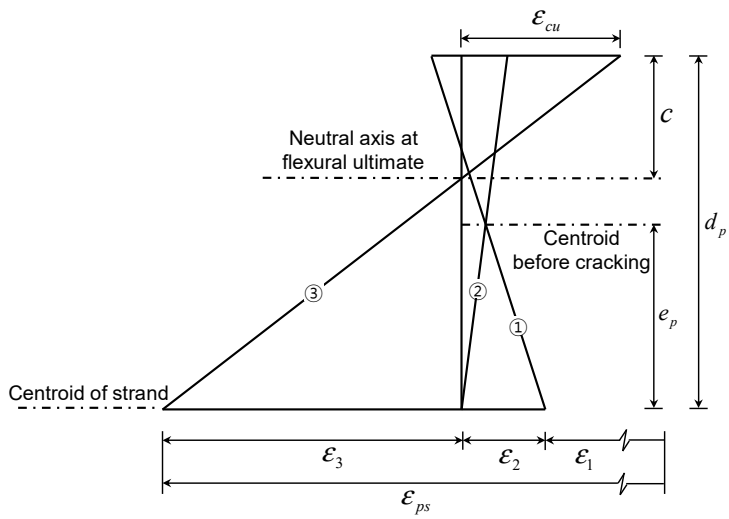

Figure 3. Strand strain at flexural failure.

$$\varepsilon_1 = \frac{f_{pe}}{E_p} = \frac{P_e}{E_p A_p}, \tag{3}$$

$$\varepsilon_2 = \frac{P_e}{E_c A_c}\left(1 + \frac{e_p^2}{r_c^2}\right), \tag{4}$$

$$\varepsilon_3 = \varepsilon_{cu}\frac{d_p - c}{c}, \tag{5}$$

$$\varepsilon_{ps} = \varepsilon_1 + \varepsilon_2 + \varepsilon_3, \tag{6}$$

where f_{pe} and P_e: effective stress and prestressing force of prestressing tendon, respectively, E_p and E_c: modulus of elasticity of prestressing tendon and concrete, respectively, where E_c is a function of f_{ck} [24], A_p and A_c: area of prestressing tendon and concrete section, respectively, e_p: eccentricity of prestressing tendon, r_c: radius of gyration of concrete section (=$\sqrt{I_c/A_c}$), I_c: second moment of area of concrete section, ε_{cu}: ultimate compressive strain of concrete, d_p: effective depth of prestressing tendon, c: distance from extreme compression fiber to neutral axis, and ε_{ps}: strain of prestressing tendon corresponding to f_{ps}.

3.4. Comparison Between Section Analysis and Existing Prediction Equations

A preliminary analysis was conducted to examine the effects of the section geometry and the strand location on f_{ps}. First, following the study by Mattock [15], a rectangular section reinforced only with strands was examined. As shown in Figure 1, section analyses were performed for representative cases of $b = 0.5h$, $b = h$, and $b = 2h$, and f_{ps} was obtained. Since the results did not show any significant differences, $b = 0.5h$ was adopted for a more detailed analysis. In addition, because the parametric analysis was conducted with respect to ρ_p, the ratio of the strand area to the concrete section area, the absolute dimensions of the section do not significantly affect the results. Therefore, in this study, b = 500 mm and h = 1000 mm were used. The effect of d_p was also examined, but since it did not lead to significant differences in the results, $d_p = 0.9h$ was assumed. Meanwhile, the validity of T-shaped and I-shaped sections was examined separately later in the study.

ρ_p was varied based on KDS [23] to ensure that sufficient section analysis results could be obtained to derive a trend line, covering both the tension-controlled region ($\varepsilon_3 \geq 0.005$) and the compression-controlled region ($\varepsilon_3 \leq 0.002$) corresponding to the strand's net tensile strain. However, in the design of PSC beam members, it is standard practice to design for a tension-controlled section that can induce ductile failure [16,23].

First, as shown in Figure 4, it was examined whether the existing prediction equation, Equation (1), is also applicable to high-strength strands. By excluding the terms related to reinforcing bar, ω and ω', in Equation (1), a linear expression with the x-axis as $\rho_p f_{pu}/f_{ck}$ and the y-axis as f_{ps}/f_{pu} was derived, and this was compared with the section analysis results in Figure 4. In this case, for β_1 included in Equation (1), a value of 0.8 was adopted, corresponding to f_{ck} = 40 MPa according to KCI [23] and KIBSE [24]. Although this differs slightly from $0.85 - 0.007(f_{ck} - 28) = 0.766$ used by Mattock [15] and ACI 318 [16], the review results are similar. For γ_p also included in Equation (1), a value of 0.28 was adopted, considering that low-relaxation strands with $f_{py}/f_{pu} \geq 0.90$ are generally used.

To quantitatively compare f_{ps} from Equation (1) with f_{ps} obtained from section analysis in Figure 4, R^2 and Mean Absolute Error (MAE) were calculated. Here, MAE is defined as in Equation (7), and an R^2 close to 1 and an MAE close to 0 indicate that the prediction equation closely matches the actual data. Since the tension-controlled region is of significant design importance, a separate MAE was also calculated for this region.

$$\text{MAE} = \sum \frac{|\hat{y}_i - y_i|}{n}, \tag{7}$$

where $\hat{y}_i$: estimated value, y_i: actual value, and n: number of samples.

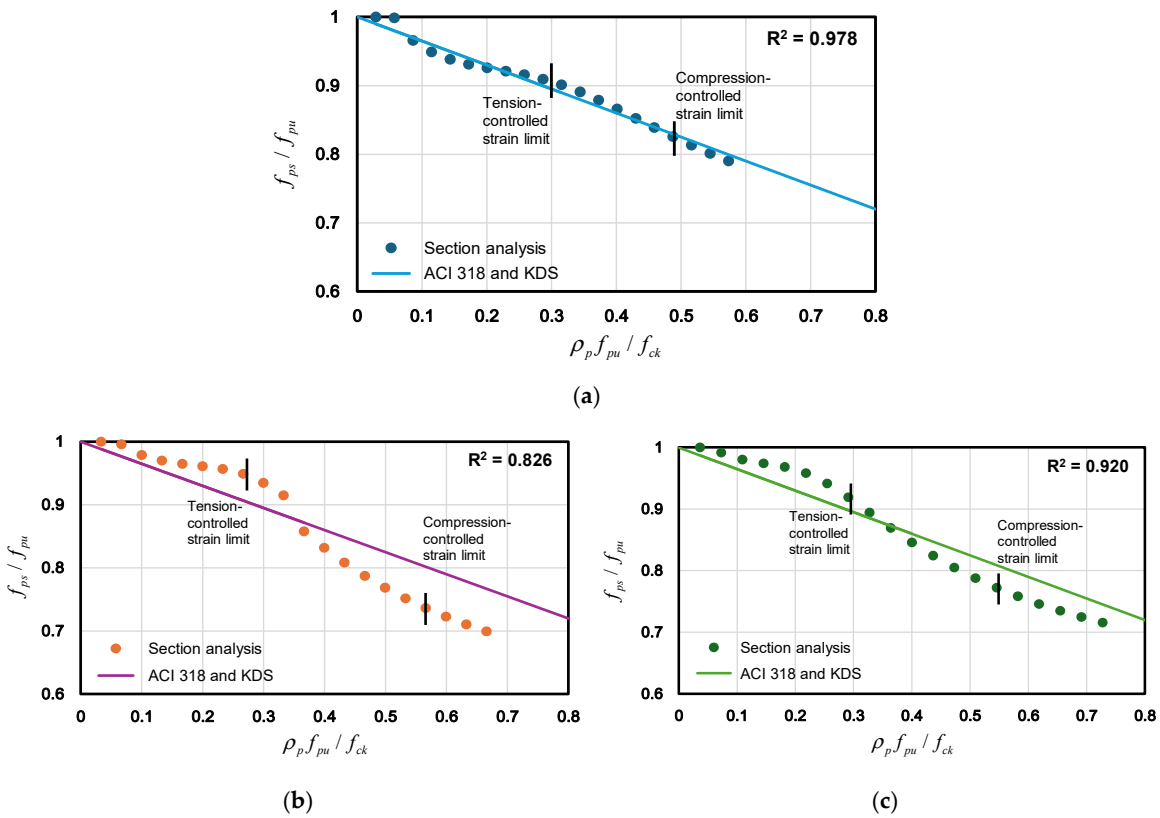

Figure 4. Comparison of existing f_{ps} prediction equations and section analysis results in each strand (rectangular section): (**a**) Type B (f_{pu} = 1860 MPa); (**b**) Type C (f_{pu} = 2160 MPa); and (**c**) Type D (f_{pu} = 2360 MPa).

According to Table 2, for Type B strands, the existing prediction equation shows a high accuracy with an R^2 value of 0.978. However, for Type C and D strands, the R^2 values are 0.826 and 0.920, respectively, indicating lower accuracy compared to Type B. Regarding MAE, Type B strands exhibited relatively small errors of 0.008 and 0.009 for the entire region and the tension-controlled region, respectively, whereas high-strength Type C and D strands showed larger errors, resulting in relatively lower accuracy of the existing prediction equation.

Table 2. Accuracy of the existing prediction equation for section analysis results.

Metric	Region	Type B (f_{pu} = 1860 MPa)	Type C (f_{pu} = 2160 MPa)	Type D (f_{pu} = 2360 MPa)
R^2	Entire	0.978	0.826	0.920
MAE	Entire	0.008	0.039	0.026
	Tension-controlled	0.009	0.025	0.024

In addition, as shown in Figure 4, the existing prediction equation exhibits high accuracy for conventional-strength Type B strands regardless of the region, whereas for high-strength Type C and D strands, the prediction trends differ depending on the region. In other words, for Type C and D strands, the existing prediction equation significantly underestimated f_{ps} in the tension-controlled region and overestimated it in the compression-controlled region. In the transition region between the tension-controlled and compression-controlled regions, the existing prediction equation also tended to overestimate f_{ps}. This

suggests that applying the existing prediction equation in the design of members with high-strength strands may be excessively conservative in the tension-controlled region, potentially reducing economic efficiency in terms of material quantities. Conversely, in the compression-controlled region and the transition region, the existing prediction equation may lead to non-conservative designs, potentially compromising structural safety. Thus, applying the existing prediction equation, which was developed for conventional-strength strands, to high-strength strands could reduce structural efficiency and safety. Therefore, in this study, the existing prediction equation was modified to more accurately predict the strand stress f_{ps} at flexural failure of PSC members with high-strength strands.

4. Proposal of an f_{ps} Prediction Equation Considering the Characteristics of High-Strength Strands

4.1. Improved f_{ps} Prediction Equation

It can be seen in Figure 4 that for the section analysis results, conventional-strength Type B strands show excellent predictive performance with the linear prediction equation. However, for high-strength Type C and D strands, there is considerable variability depending on the region, and a strong tendency to deviate from linearity. This tendency was particularly more pronounced in Type C than in Type D. Considering this, it would be possible to propose new prediction equations for Types C and D based on trend lines of a curved shape; however, there are doubts regarding the practicality of this approach in terms of ease of use in engineering practice.

Therefore, in this study, considering practical convenience, the basic linear form of the f_{ps} prediction equation in Equation (1), as adopted in ACI 318 [16] and KDS [14], was retained, while factors reflecting the characteristics of high-strength strands were added to improve the predictive equation. Based on the analysis of various trend line forms, the proposed f_{ps} prediction equation for high-strength strands is given in Equation (8).

$$f_{ps} = \alpha f_{pu}\left\{1 - k\frac{\gamma_p}{\beta_1}\left(\rho_p\frac{f_{pu}}{f_{ck}}\right)\right\}, \tag{8}$$

where α: 1.00, 1.01, and 1.02 for f_{pu} = 1860 MPa, 2160 MPa, and 2360 MPa, respectively, and $k = f_{pu}/1860$. The correction terms α and k incorporate the influence of the strand tensile strength. When compared with Figure 4b,c, α adjusts both the y-intercept and the slope compared with the existing prediction equation, while k serves to modify the slope. If only the slope of the prediction equation is adjusted while keeping the intercept fixed at 1, as in the existing equation, the shapes observed in Figure 4b,c indicate that the prediction accuracy in certain regions would improve, whereas the accuracy in other regions would decrease. To mitigate this tendency, it is desirable to adjust the intercept as well, which resulted in the form shown in Equation (8). In addition, when Equation (8) is applied to conventional-strength strands with f_{pu} = 1860 MPa, it yields the same form as Equation (1), indicating that consistency with the existing equation is maintained.

Meanwhile, in Equation (1) originally proposed by Mattock [15], the term $(d/d_p)(\omega - \omega')$ is included to encompass cases in PSC member design where cracking under service loads is permitted and such cracks are controlled by reinforcing bars. This accounts for the redistribution of stresses after cracking due to the effects of tension and compression reinforcing bars, which causes the magnitude of f_{ps} to vary. However, since this study focused on improving the prediction accuracy of f_{ps} by reflecting the characteristics of high-strength strands, no separate evaluation was conducted for the terms accounting for reinforcing bars.

Until a separate study on this is conducted, it is recommended to use Equation (8) with the existing reinforcing-bars-related terms additionally considered, as shown in Equation (9).

$$f_{ps} = \alpha f_{pu}\left[1 - k\frac{\gamma_p}{\beta_1}\left\{\rho_p\frac{f_{pu}}{f_{ck}} + \frac{d}{d_p}(\omega - \omega')\right\}\right], \tag{9}$$

4.2. Evaluation of the Proposed Equation

To evaluate how much more accurately the proposed f_{ps} prediction equation can estimate the strand stress at flexural failure in members using high-strength strands compared to the existing equation, it was compared with the results of various section analyses. Specifically, precise section analyses were performed not only for rectangular sections but also for T-shaped and I-shaped sections to calculate f_{ps}, and the results were then compared with f_{ps} values obtained using the existing and proposed formulas, corresponding to Equations (1) and (8), respectively.

In addition, by calculating the nominal flexural strength, which includes f_{ps}, and comparing the results, the performance of the proposed equation in predicting the flexural strength of members relative to the existing equation was examined. In this way, the suitability of the proposed equation was evaluated by assessing its accuracy in predicting both the strand stress and the flexural strength at flexural failure for members with high-strength strands.

The T-shaped section additionally examined was partly based on the cross-sectional geometry used in Mattock's study [15], as shown in Figure 5a, and was configured with $h = b = 2000$ mm and $d_p = (h - 100)$ mm. Meanwhile, for the I-shaped section shown in Figure 5b, the cross-sectional geometry of the standard 30-m-span PSC girder proposed by the Korea Expressway Corporation was examined. However, since the parametric analysis was conducted with respect to ρ_p, as in the rectangular section analysis, the influence of the absolute dimensions of the section is not significant. Other than that, the material models and analysis procedures for the section analysis are identical to those used for the preceding rectangular section.

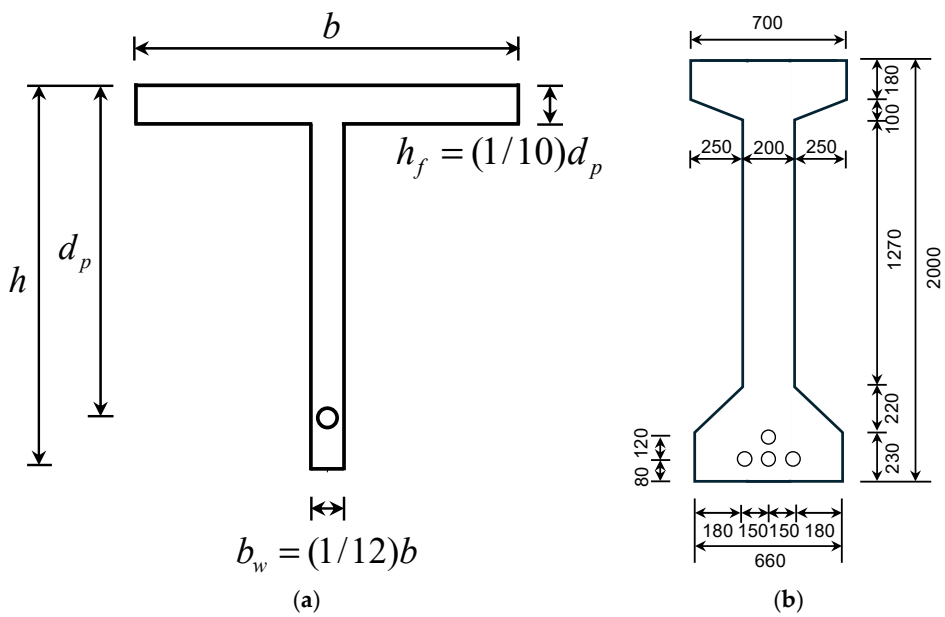

Figure 5. T-shaped and I-shaped sections used for the section analysis: (**a**) T-shaped section; and (**b**) I-shaped section.

The proposed and the existing equations for f_{ps} were compared with the section analysis results, as shown in Figure 6. By adjusting the y-intercept and slope of the existing equations, the proposed equation improved the prediction accuracy relative to the section analysis results for all section types—rectangular, T-shaped, and I-shaped—and demonstrated particularly enhanced performance in the tension-controlled region, which is critical in practical design. For the rectangular section, when using the proposed equation, the R^2 values improved compared to the existing equations—from 0.826 to 0.900 for Type C and from 0.920 to 0.952 for Type D.

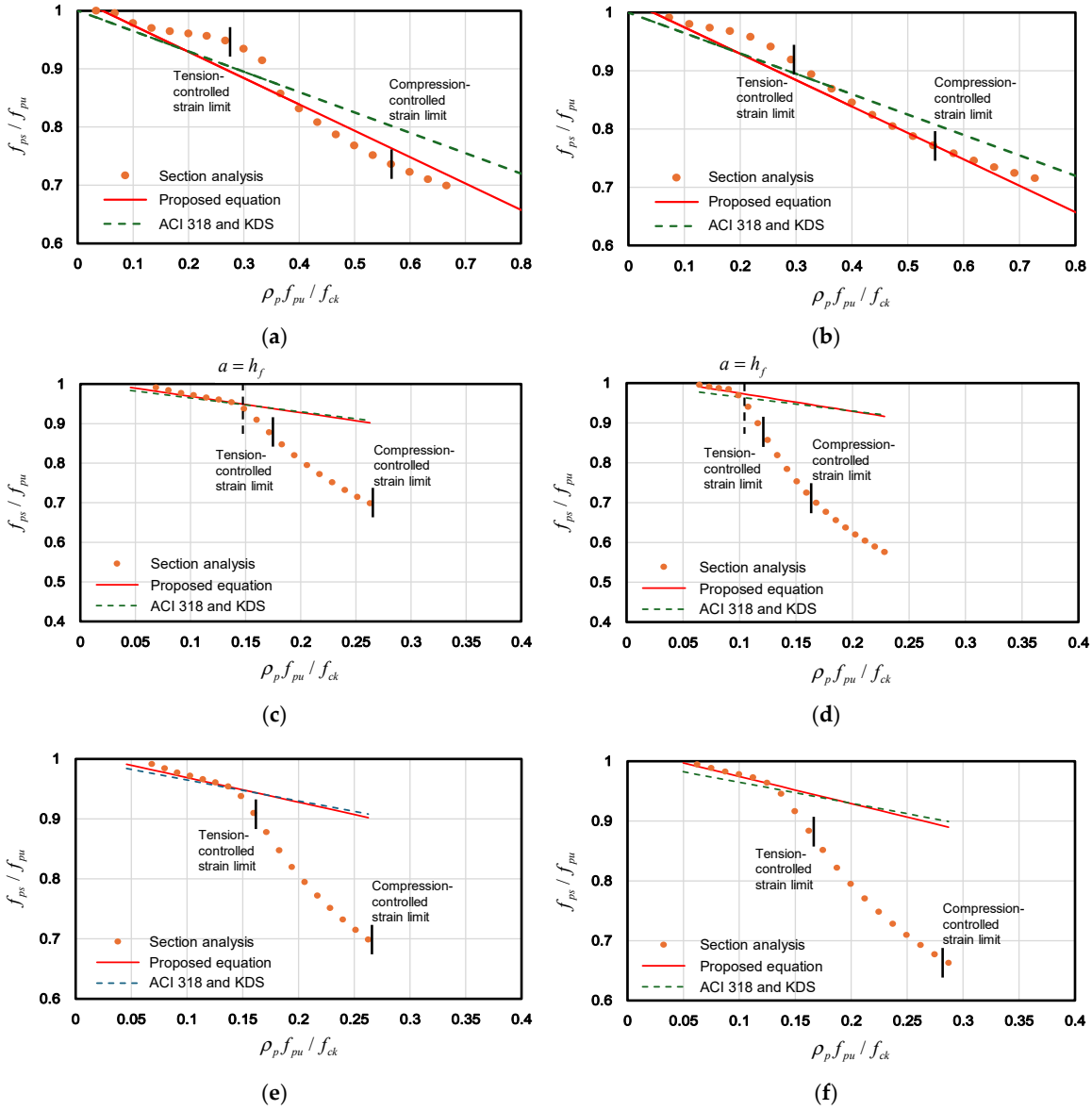

Figure 6. Comparison of f_{ps} predicted by the equation and section analysis results for high-strength strands in different section shapes: (**a**) rectangular section, Type C (f_{pu} = 2160 MPa); (**b**) rectangular section, Type D (f_{pu} = 2360 MPa); (**c**) T-shaped section, Type C (f_{pu} = 2160 MPa); (**d**) T-shaped section, Type D (f_{pu} = 2360 MPa); (**e**) I-shaped section, Type C (f_{pu} = 2160 MPa); and (**f**) I-shaped section, Type D (f_{pu} = 2360 MPa).

Meanwhile, since both the existing and proposed equations were developed based on the rectangular section analysis, the proposed equation still provides reasonable prediction performance for T-shaped and I-shaped sections when the neutral axis is located in the top flange, as their behavior is similar to that of the rectangular section. However, when the

neutral axis extends beyond the top flange into the web, the equations gradually deviate from a reasonable prediction range. Figure 6c,d illustrate the case for the T-shaped section in Figure 5a where a from Figure 1d, calculated from the neutral axis, is located at the bottom of the top flange ($a = h_f$). As mentioned in Section 2, this issue was also pointed out in Mattock's study [15], who addressed it by limiting the amount of prestressing tendon. Examining the results for the T-shaped and I-shaped sections in Figure 6c–f, although the tension-controlled region does not exactly coincide with the region where the neutral axis is located in the top flange, they are generally similar. Therefore, the proposed equation remains effective, as it significantly improves prediction accuracy in the tension-controlled region, which is a standard design practice for inducing a desirable ductile failure mode, for both Type C and Type D strands.

f_{ps} can be used to estimate the flexural resistance of PSC member sections, and, for example, the nominal flexural strength according to the ultimate strength design method can be calculated from Equation (10) based on Figure 1c and from Equation (11) based on Figure 1d. According to KDS [23,24], when f_{ck} = 40 MPa, Figure 1 gives α = 0.8, β = 0.4, and η = 1.0. For a rectangular section, Equations (10) and (11) produce identical results. As observed in Equations (10) and (11), f_{ps} is associated with the tensile force of the strands, and thus it significantly affects the depth of the neutral axis and the nominal flexural strength.

$$M_n = A_p f_{ps}(d_p - \beta c) \text{ where } c = \frac{A_p f_{ps}}{\alpha(0.85 f_{ck})b}, \tag{10}$$

$$M_n = A_p f_{ps}\left(d_p - \frac{a}{2}\right) \text{ where } a = \frac{A_p f_{ps}}{\eta(0.85 f_{ck})b}, \tag{11}$$

Figure 7 illustrates the difference between using the proposed f_{ps} equation and the existing equation when calculating the nominal flexural strength using Equation (10) or (11). On the y-axis in Figure 7, $M_{n,SA}$ represents the nominal flexural strength calculated using f_{ps} obtained from the section analysis, while $M_{n,code}$ is calculated using the predictive equations for f_{ps}, where the existing equation in Equation (1) and the proposed equation in Equation (8) are applied separately. When $M_{n,code}/M_{n,SA}$ is less than 1, the result can be regarded as conservative, whereas a value greater than 1 indicates a non-conservative outcome. From a design perspective, it is desirable for the prediction equation for f_{ps} to yield conservative results to ensure structural safety.

As observed in Figure 7, regardless of the section shape or the prediction equation used, the tensile-controlled region generally exhibits a conservative tendency, whereas the transition and compression-controlled regions tend to be non-conservative. However, since these non-conservative regions may potentially lead to brittle failure, they are avoided in actual beam design, and thus this does not pose a significant problem. The analysis results showed that when the proposed f_{ps} equation was used, conservativeness in the tensile-controlled region was maintained, and the prediction accuracy improved across all regions, with $M_{n,code}/M_{n,SA}$ approaching 1 compared to the existing equation. In particular, when Type D strands were used in a rectangular section, the proposed equation adjusted $M_{n,code}/M_{n,SA}$ to be less than or equal to 1 even in the transition and compression-controlled regions, resulting in conservative outcomes across all regions.

In this way, the proposed equation appropriately reflects the material characteristics of high-strength strands, thereby improving the prediction accuracy of strand stress f_{ps} and nominal flexural strength M_n at the ultimate state during beam flexural failure. As a result, it is expected to ensure an adequate level of safety while also considering material efficiency.

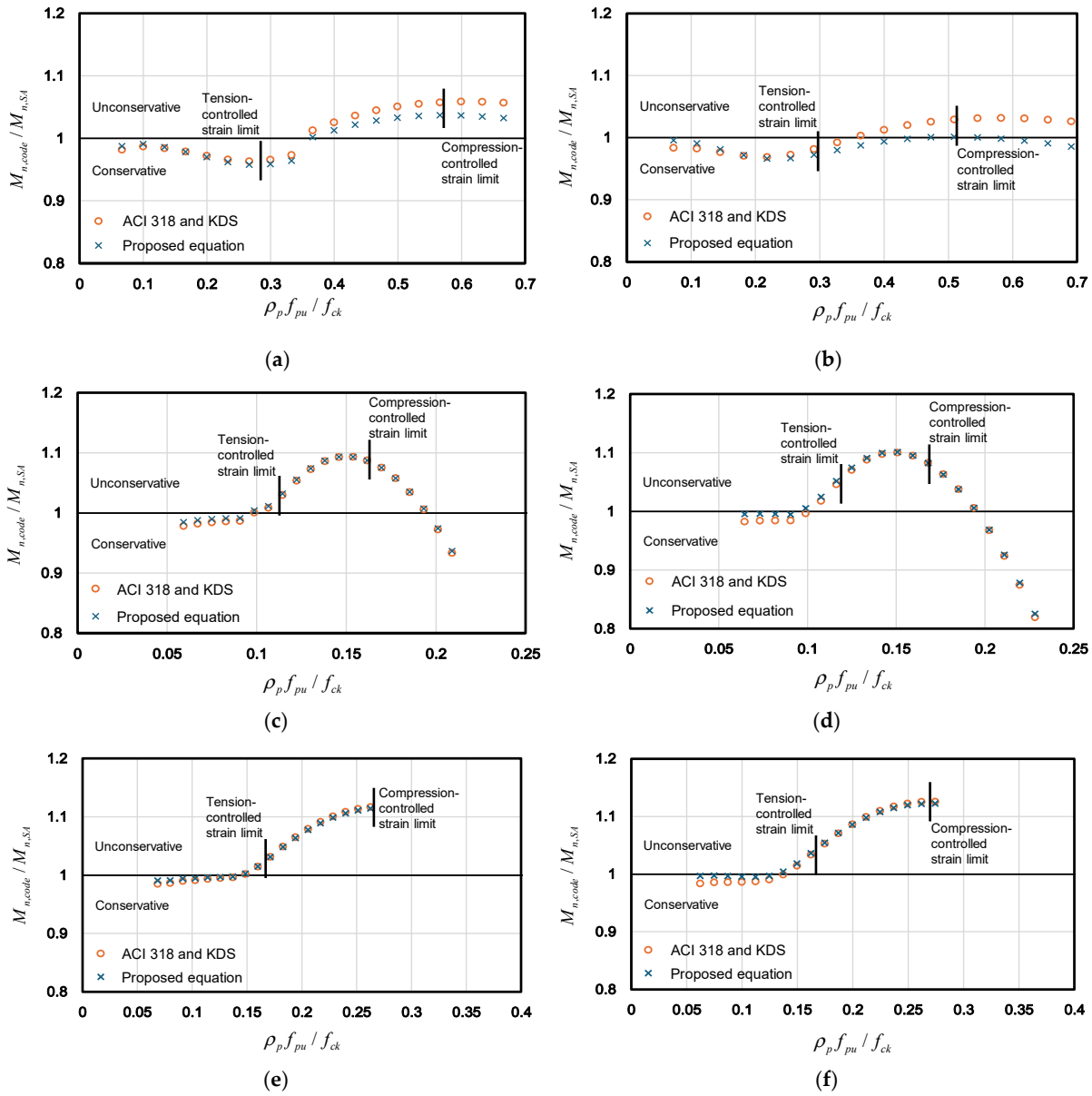

Figure 7. Comparison of M_n based on f_{ps} of high-strength strands for different section shapes: (**a**) rectangular section, Type C (f_{pu} = 2160 MPa); (**b**) rectangular section, Type D (f_{pu} = 2360 MPa); (**c**) T-shaped section, Type C (f_{pu} = 2160 MPa); (**d**) T-shaped section, Type D (f_{pu} = 2360 MPa); (**e**) I-shaped section, Type C (f_{pu} = 2160 MPa); and (**f**) I-shaped section, Type D (f_{pu} = 2360 MPa).

5. Conclusions

In this study, to ensure both applicability and safety when using high-strength strands in PSC structures, the suitability of existing prediction equations for strand stress f_{ps} at beam flexural failure was evaluated, and an improved prediction equation was proposed. The analysis results can be summarized as follows.

1. The existing f_{ps} prediction equations presented in ACI 318 and KDS were derived from studies on conventional strands with a tensile strength of 1860 MPa. By establishing stress–strain curves for each strand and performing rigorous section analysis corresponding to the exact solution, it was confirmed that these existing equations can predict the strand stress with high accuracy for 1860 MPa strands. However, it was also confirmed that the stress prediction performance of the existing equations deteriorates for high-strength strands with tensile strengths of 2160 and 2360 MPa. Specifically, compared to conventional-strength strands, the existing equations tend

to underestimate the stress in the tensile-controlled region—commonly adopted in most designs—thereby promoting ductile failure in PSC beams, while overestimating the stress in the transition and compression-controlled regions. Therefore, for high-strength strands, it is considered more appropriate to develop an improved f_{ps} equation rather than directly applying the existing equations.

2. The improved f_{ps} prediction equation was derived through section analyses based on highly accurate mathematical expressions obtained from tensile test data of multiple high-strength strands. To ensure practicality, correction terms were introduced to the f_{ps} prediction equation for 1860 MPa strands presented in ACI 318 and KDS, reflecting the increased tensile strength characteristics of 2160 and 2360 MPa high-strength strands, thereby improving the agreement with section analysis results. When applying the proposed equation to rectangular sections, the coefficient of determination (R^2) increased from 0.826 to 0.900 for 2160 MPa strands and from 0.920 to 0.952 for 2360 MPa strands, demonstrating improved prediction performance compared to the existing equation. Since the prediction equation was derived from the analysis of rectangular sections, its accuracy inevitably decreases for T-shaped or I-shaped sections with top flanges once the neutral axis extends beyond the top flange. However, for tensile-controlled sections, which correspond to most PSC beam designs, the prediction accuracy was found to be closer to the section analysis results, indicating enhanced predictive performance.
3. In the review of nominal flexural strength based on the ultimate strength design method, it was also found that using the proposed f_{ps} prediction equation improves the prediction performance compared to the existing equation. Specifically, when the proposed equation is used, conservativeness is maintained in the tensile-controlled region, while the nominal flexural strength is predicted more accurately, closely matching the results of the section analysis compared to the existing equation.
4. Recently, high-strength strands with tensile strengths of 2160 and 2360 MPa have been additionally included in ISO standards, and the application of high-strength strands in infrastructure structures is increasing worldwide. For high-strength strands to be more widely utilized, it is necessary to verify the validity of design equations originally proposed for conventional-strength strands and, if needed, to develop improved equations more suitable for high-strength strands. In this study, a rationally improved equation was proposed specifically for the strand stress at beam failure, which is the most critical aspect of PSC beam flexural design. It is expected that further research and incorporation into design codes will be required for other related design equations in the future.
5. Since high-strength strands have a higher yield strain than conventional-strength strands, the net tensile strain limit of strands defining a tension-controlled section for inducing ductile failure may be greater than 0.005. This issue will be clarified through a ductility analysis in future studies.

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